



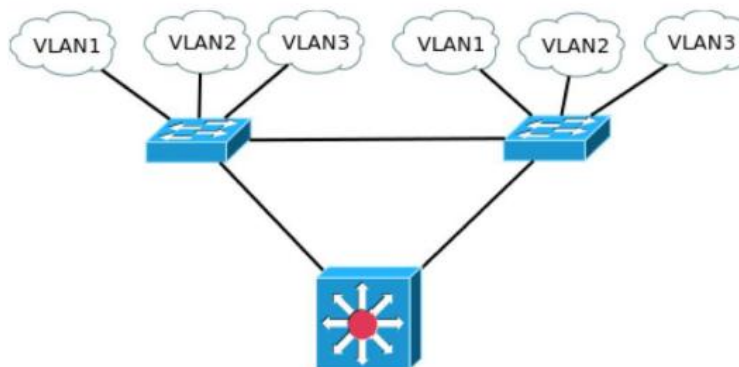
Guia 5– Interligação de VLANs

Objectivos:

- Routing entre VLANs
- Estabelecimento e configuração de trunks

1 - Inter-VLAN Routing: usando um Switch L3 (e links redundantes)

5. Troque o router por um switch L3 conforme a figura abaixo. Tenha em atenção quais as interfaces que usa para ligar o switch L3 aos outros!



Configure 3 VLANs no L3 Switch (VLAN1 ,2 e 3):

```
RouterSW# vlan database
RouterSW(vlan)# vlan 1
RouterSW(vlan)# vlan 2
RouterSW(vlan)# vlan 3
RouterSW(vlan)# exit
```

Configure os portos L2 do Switch L3 (fastEthernet slot 1/*), porto 0: VLAN1, portos 1-8: VLAN2, portos 9-12:

VLAN3 e portos 13-15: Inter-switch/Tagged/802.1Q:

```
RouterSW(config)# interface f1/0
RouterSW(config-if)# switchport mode access
RouterSW(config-if)# switchport access vlan 1
RouterSW(config-if)# interface range fastEthernet 1/1 - 8
RouterSW(config-if-range)# switchport mode access
RouterSW(config-if-range)# switchport access vlan 2
RouterSW(config-if-range)# interface range fastEthernet 1/9 - 12
RouterSW(config-if-range)# switchport mode access
```

Cofinanciado por:





```
RouterSW(config-if-range)# switchport access vlan 3
RouterSW(config-if-range)# interface range fastEthernet 1/13 - 15
RouterSW(config-if-range)# switchport mode trunk
RouterSW(config-if-range)# switchport trunk encapsulation dot1q

Configure as interfaces virtuais (Vlan) do switch L3:
RouterSW(config)# interface Vlan 1
RouterSW(config-if)# ip address 10.1.1.1 255.255.255.0
RouterSW(config-if)# no autostate !força o porto a estar sempre em cima

RouterSW(config)# interface Vlan 2
RouterSW(config-if)# ip address 10.2.2.1 255.255.255.0
RouterSW(config-if)# no autostate ! força o porto a estar sempre em cima
RouterSW(config)# interface Vlan 3
RouterSW(config-if)# ip address 10.3.3.1 255.255.255.0
RouterSW(config-if)# no autostate !forces the port to be always up
```

Verifique a tabela de encaminhamento (**show ip route**). Certifique-se que os terminais (hosts) têm as gateways configuradas para os endereços IP das interfaces virtuais (Vlan) do switch L3 e teste a conectividade. Capture os pacotes trocados entre o Router e o Switch L3.

6. Grave as configurações dos equipamentos, bem como o projecto.

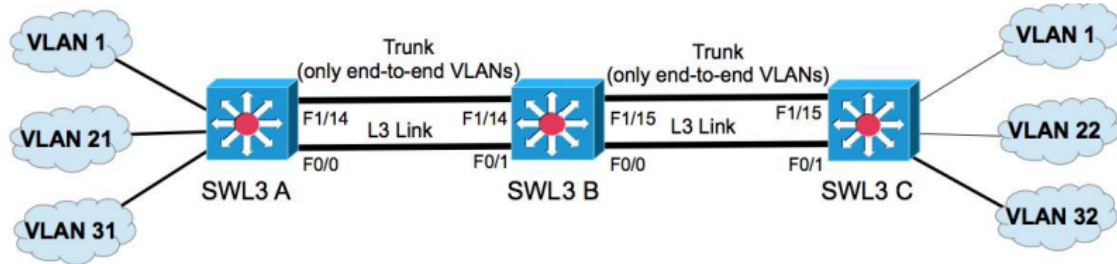
1 – Links de Trunk Restringidos

7. Construa a seguinte rede usando Switches L3. A VLAN 1 existe de uma ponta à outra da rede (end-to-end), e as outras VLANs são apenas locais. Para simular interfaces L3, use interfaces Ethernet normais, ou interfaces L2 (modulo de Switching) convertidas em interfaces L3 usando o commando no switchport.

Configure as VLANs nos três switches L3: SWL3 A deverá ter as VLAN 1, 21, and 31; SWL3 B deverá ter a VLAN 1 apenas; e o SWL3 C deverá ter as VLAN 1, 22, and 32.

Nota: Por defeito, os Cisco têm VLANs que têm que ser consideradas end-to-end (1002-1005) e não podem ser apagadas.

Nota: Não se esqueça que pode substituir as núvens por VPC's ou outros routers a fazer de host.



Configure as VLANs assumindo que os portos de trunk são f1/14 e f1/15 em todos os switches L3:

```
SWL3*(config-if-range)# interface range fastEthernet 1/14 - 15
```

```
SWL3*(config-if-range)# switchport mode trunk
```

```
SWL3*(config-if-range)# switchport trunk allowed vlan 1,1002-1005
```

Garanta que todos os switches L3 têm o routing IPv4 ativo, usando o comando `ip routing`.

Configure as os endereços IP das interfaces assumindo que as VLANs têm as redes 10.0.<#vlan>.0/24, a rede IP entre o SWL3A e o SWL3B é a 10.1.0.0/24 e a rede entre o SWL3B e o SWL3C é a 10.2.0.0/24.

Configure um protocolo de encaminhamento.

```
SWL3*(config)# router rip
```

```
SWL3*(config-router)# version 2 !Necessário devido às sub-redes
```

```
SWL3*(config-router)# network 10.0.0.0
```

```
SWL3*(config-router)# passive-interface vlan 1 ! Não utiliza a VLAN1 para encaminhar tráfego L3
```

Analise as tabelas de encaminhamento, teste a conectividade e capture/analise as capturas no trunk e nos links L3.

Nota:

Para fazer um ping desde a interface da VLAN1 no SWL3A (e.g., 10.0.1.1) para a interface VLAN22 do SWL3C (e.g., 10.0.22.3), utilize o comando ping com indicação da origem:

```
SWL3A# ping 10.0.22.3 source 10.0.1.1
```

Sem o comando `source` o endereço origem do ping é o da interface de saída definida pela tabela de encaminhamento.

Conclua acerca de que tráfego passa através de que link, entre os switches, quando existe conectividade inter-VLAN.

8. Grave as configurações dos equipamentos, bem como o projecto.

2 – Links de Trunk Restringidos (sem Link L3)

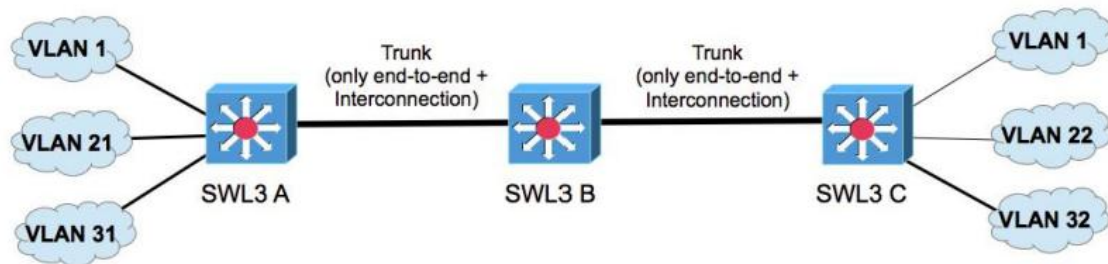
9. Remova o link L3, crie uma VLAN de interligação (e.g., VLAN 101) e reconfigure a rede para ter conectividade total entre VLANs. Comece por alterar as restrições do trunk:

```
SWL3*(config-if-range)# switchport trunk allowed vlan 1,101,1002-1005
```



Analise as tabelas de encaminhamento, teste a conectividade e capture/analise as capturas nos trunks.

DICA: As VLANs de interligação podem ser usadas para passar anúncios de rotas RIP de outras redes.

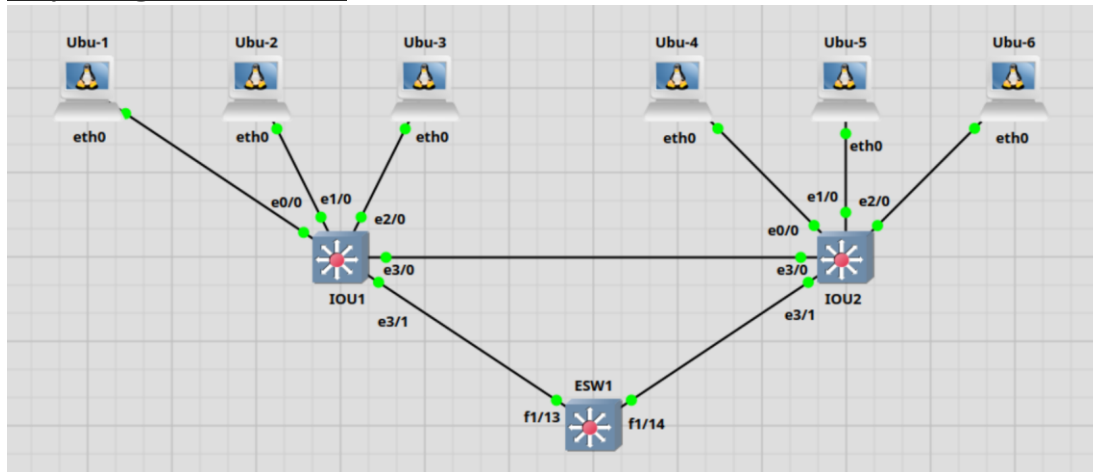


10. Grave as configurações dos equipamentos, bem como o projecto.

Inter-VLAN Routing: usando um Switch L3 (e links redundantes)

5.

Topologia da rede:



Configuração do Switch de Layer 3 ESW1:

```
ESW1#sh run
```

```
Building configuration...
```

```
Current configuration : 3096 bytes
```

```
!
```

```
version 12.4
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
no service password-encryption
```

```
no service dhcp
```

```
!
```

```
hostname ESW1
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!
```

```
no aaa new-model
```

```
memory-size iomem 5
```

```
no ip icmp rate-limit unreachable
```

```
ip cef
```

```
!  
!  
!  
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100  
!  
interface FastEthernet1/1
```

```
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/2
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/3
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/4
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/5
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/6
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/7
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/8
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/9
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/10
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/11
switchport access vlan 3
```

```
duplex full
speed 100
!
interface FastEthernet1/12
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/13
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/14
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
ip address 10.1.1.1 255.255.255.0
no autostate
!
interface Vlan2
ip address 10.2.2.1 255.255.255.0
no autostate
!
interface Vlan3
ip address 10.3.3.1 255.255.255.0
no autostate
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
```

```
!  
control-plane
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
banner exec ^C
```

```
*****
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
*****
```

```
^C
```

```
!
```

```
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line vty 0 4  
  login  
!  
!  
end
```

Há conectividade entre todas as VLANs:

```
root@Ubu-1:~# ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=64 time=0.020 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=64 time=0.055 ms
^C
--- 10.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 0.020/0.037/0.055/0.017 ms
root@Ubu-1:~# ping 10.1.1.3
PING 10.1.1.3 (10.1.1.3) 56(84) bytes of data.
64 bytes from 10.1.1.3: icmp_seq=1 ttl=64 time=7.57 ms
64 bytes from 10.1.1.3: icmp_seq=2 ttl=64 time=1.58 ms
64 bytes from 10.1.1.3: icmp_seq=3 ttl=64 time=2.18 ms
64 bytes from 10.1.1.3: icmp_seq=4 ttl=64 time=10.8 ms
64 bytes from 10.1.1.3: icmp_seq=5 ttl=64 time=1.65 ms
64 bytes from 10.1.1.3: icmp_seq=6 ttl=64 time=1.54 ms
^C
--- 10.1.1.3 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 1.537/4.213/10.756/3.625 ms
root@Ubu-1:~# ping 10.2.2.3
PING 10.2.2.3 (10.2.2.3) 56(84) bytes of data.
64 bytes from 10.2.2.3: icmp_seq=2 ttl=63 time=16.9 ms
64 bytes from 10.2.2.3: icmp_seq=3 ttl=63 time=14.6 ms
64 bytes from 10.2.2.3: icmp_seq=4 ttl=63 time=19.0 ms
^C
--- 10.2.2.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3037ms
rtt min/avg/max/mdev = 14.601/16.823/19.008/1.799 ms
root@Ubu-1:~# ping 10.3.3.3
PING 10.3.3.3 (10.3.3.3) 56(84) bytes of data.
64 bytes from 10.3.3.3: icmp_seq=2 ttl=63 time=21.4 ms
64 bytes from 10.3.3.3: icmp_seq=3 ttl=63 time=22.3 ms
64 bytes from 10.3.3.3: icmp_seq=4 ttl=63 time=19.0 ms
^C
--- 10.3.3.3 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3043ms
rtt min/avg/max/mdev = 18.974/20.878/22.300/1.399 ms
ESW1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 3 subnets
C       10.3.3.0 is directly connected, Vlan3
C       10.2.2.0 is directly connected, Vlan2
C       10.1.1.0 is directly connected, Vlan1
```

Configuração do switch Layer 2, IOU1:

IOU1#sh run

Building configuration...

Current configuration : 2124 bytes

!

! Last configuration change at 21:01:54 WET Fri Feb 27 2026

!

version 15.2

service timestamps debug datetime msec

```
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU1
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
clock timezone WET 0 0
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
```

```
switchport mode access
!
interface Ethernet0/1
switchport mode access
!
interface Ethernet0/2
switchport mode access
!
interface Ethernet0/3
switchport mode access
!
interface Ethernet1/0
switchport access vlan 2
switchport mode access
!
interface Ethernet1/1
switchport access vlan 2
switchport mode access
!
interface Ethernet1/2
switchport access vlan 2
switchport mode access
!
interface Ethernet1/3
switchport access vlan 2
switchport mode access
!
interface Ethernet2/0
switchport access vlan 3
switchport mode access
!
interface Ethernet2/1
switchport access vlan 3
switchport mode access
!
interface Ethernet2/2
switchport access vlan 3
switchport mode access
!
interface Ethernet2/3
switchport access vlan 3
switchport mode access
!
interface Ethernet3/0
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
```

```

interface Ethernet3/2
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/3
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end

```

Configuração do switch Layer 2, IOU2:

```

IOU2#sh run
Building configuration...

Current configuration : 2124 bytes
!
! Last configuration change at 21:01:54 WET Fri Feb 27 2026
!
version 15.2
service timestamps debug datetime msec

```

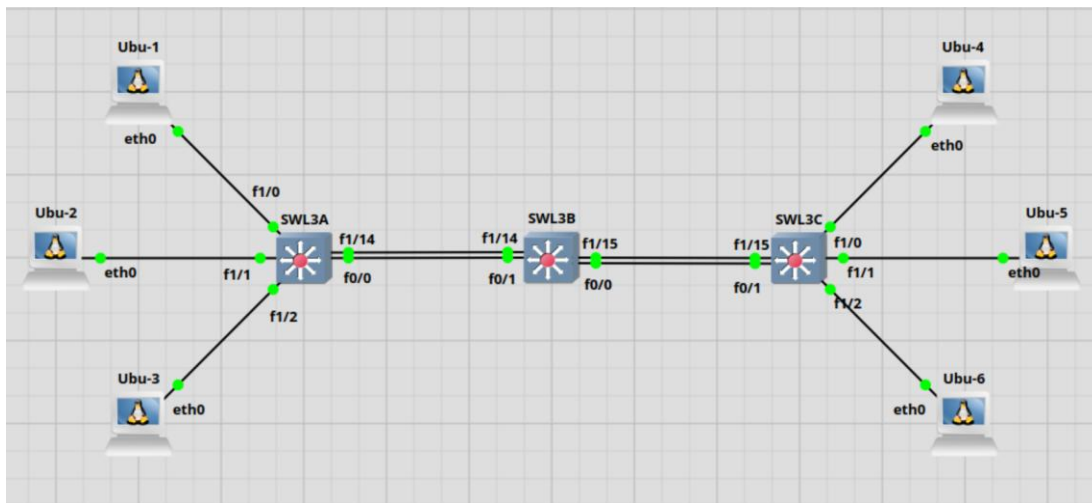
```
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname IOU2
!
boot-start-marker
boot-end-marker
!
!
logging discriminator EXCESS severity drops 6 msg-body drops EXCESSCOLL
logging buffered 50000
logging console discriminator EXCESS
!
no aaa new-model
clock timezone WET 0 0
!
!
!
!
no ip icmp rate-limit unreachable
!
!
!
no ip domain-lookup
ip cef
no ipv6 cef
!
!
!
spanning-tree mode rapid-pvst
spanning-tree extend system-id
!
vlan internal allocation policy ascending
!
ip tcp synwait-time 5
!
!
!
!
!
!
!
!
!
!
!
interface Ethernet0/0
```

```
switchport mode access
!
interface Ethernet0/1
switchport mode access
!
interface Ethernet0/2
switchport mode access
!
interface Ethernet0/3
switchport mode access
!
interface Ethernet1/0
switchport access vlan 2
switchport mode access
!
interface Ethernet1/1
switchport access vlan 2
switchport mode access
!
interface Ethernet1/2
switchport access vlan 2
switchport mode access
!
interface Ethernet1/3
switchport access vlan 2
switchport mode access
!
interface Ethernet2/0
switchport access vlan 3
switchport mode access
!
interface Ethernet2/1
switchport access vlan 3
switchport mode access
!
interface Ethernet2/2
switchport access vlan 3
switchport mode access
!
interface Ethernet2/3
switchport access vlan 3
switchport mode access
!
interface Ethernet3/0
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/1
switchport trunk encapsulation dot1q
switchport mode trunk
!
```

```
interface Ethernet3/2
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Ethernet3/3
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Vlan1
no ip address
shutdown
!
ip forward-protocol nd
!
no ip http server
no ip http secure-server
!
!
!
!
!
control-plane
!
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end
```

Links de Trunk Restringidos

7.



Configuração do SWL3A:

```
SWL3A#sh run
```

```
Building configuration...
```

```
Current configuration : 2982 bytes
```

```
!
```

```
version 12.4
```

```
service timestamps debug datetime msec
```

```
service timestamps log datetime msec
```

```
no service password-encryption
```

```
no service dhcp
```

```
!
```

```
hostname ESW1
```

```
!
```

```
boot-start-marker
```

```
boot-end-marker
```

```
!
```

```
!
```

```
no aaa new-model
```

```
memory-size iomem 5
```

```
no ip icmp rate-limit unreachable
```

```
ip cef
```

```
!
```

```
!
```

```
!
```

```
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.1.0.2 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100
```

```
!  
interface FastEthernet1/1  
  switchport access vlan 21  
  duplex full  
  speed 100  
!  
interface FastEthernet1/2  
  switchport access vlan 31  
  duplex full  
  speed 100  
!  
interface FastEthernet1/3  
  duplex full  
  speed 100  
!  
interface FastEthernet1/4  
  duplex full  
  speed 100  
!  
interface FastEthernet1/5  
  duplex full  
  speed 100  
!  
interface FastEthernet1/6  
  duplex full  
  speed 100  
!  
interface FastEthernet1/7  
  duplex full  
  speed 100  
!  
interface FastEthernet1/8  
  duplex full  
  speed 100  
!  
interface FastEthernet1/9  
  duplex full  
  speed 100  
!  
interface FastEthernet1/10  
  duplex full  
  speed 100  
!  
interface FastEthernet1/11  
  duplex full
```

```
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
ip address 10.0.1.1 255.255.255.0
no autostate
!
interface Vlan21
ip address 10.0.21.1 255.255.255.0
no autostate
!
interface Vlan31
ip address 10.0.31.1 255.255.255.0
no autostate
!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
```

```
ip forward-protocol nd
!  
!  
no ip http server  
no ip http secure-server  
!  
no cdp log mismatch duplex  
!  
!  
!  
!  
control-plane  
!  
!  
!  
!  
!  
!  
!  
!  
!  
banner exec ^C
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
^C  
!  
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line aux 0  
  exec-timeout 0 0  
  privilege level 15
```

```
logging synchronous
line vty 0 4
login
!
!
end
```

```
ESW1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 7 subnets
R       10.2.0.0 [120/1] via 10.1.0.1, 00:00:09, FastEthernet0/0
C       10.1.0.0 is directly connected, FastEthernet0/0
C       10.0.1.0 is directly connected, Vlan1
C       10.0.31.0 is directly connected, Vlan31
R       10.0.22.0 [120/2] via 10.1.0.1, 00:00:09, FastEthernet0/0
C       10.0.21.0 is directly connected, Vlan21
R       10.0.32.0 [120/2] via 10.1.0.1, 00:00:09, FastEthernet0/0
```

Configuração do SWL3B:

```
SWL3B#sh run
Building configuration...

Current configuration : 2776 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname ESW2
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
```

```
!  
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.2.0.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.1.0.1 255.255.255.0  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100
```

```
!  
interface FastEthernet1/1  
  duplex full  
  speed 100  
!  
interface FastEthernet1/2  
  duplex full  
  speed 100  
!  
interface FastEthernet1/3  
  duplex full  
  speed 100  
!  
interface FastEthernet1/4  
  duplex full  
  speed 100  
!  
interface FastEthernet1/5  
  duplex full  
  speed 100  
!  
interface FastEthernet1/6  
  duplex full  
  speed 100  
!  
interface FastEthernet1/7  
  duplex full  
  speed 100  
!  
interface FastEthernet1/8  
  duplex full  
  speed 100  
!  
interface FastEthernet1/9  
  duplex full  
  speed 100  
!  
interface FastEthernet1/10  
  duplex full  
  speed 100  
!  
interface FastEthernet1/11  
  duplex full  
  speed 100  
!
```

```
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
```

```
!  
control-plane
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
banner exec ^C
```

```
*****
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
*****
```

```
^C
```

```
!
```

```
line con 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line aux 0  
  exec-timeout 0 0  
  privilege level 15  
  logging synchronous  
line vty 0 4  
  login  
!  
!  
end
```

```

ESW2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 7 subnets
C       10.2.0.0 is directly connected, FastEthernet0/0
C       10.1.0.0 is directly connected, FastEthernet0/1
R       10.0.1.0 [120/1] via 10.2.0.2, 00:00:09, FastEthernet0/0
          [120/1] via 10.1.0.2, 00:00:25, FastEthernet0/1
R       10.0.31.0 [120/1] via 10.1.0.2, 00:00:25, FastEthernet0/1
R       10.0.22.0 [120/1] via 10.2.0.2, 00:00:09, FastEthernet0/0
R       10.0.21.0 [120/1] via 10.1.0.2, 00:00:25, FastEthernet0/1
R       10.0.32.0 [120/1] via 10.2.0.2, 00:00:10, FastEthernet0/0

```

Configuração do SWL3C:

SWL3C#sh run

Building configuration...

Current configuration : 2982 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

no service dhcp

!

hostname ESW3

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip icmp rate-limit unreachable

ip cef

!

!

!

!

no ip domain lookup

ip auth-proxy max-nodata-conns 3

```
ip admission max-nodata-conns 3
```

```
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!
```

```
vtp file nvram:vlan.dat
```

```
!  
!
```

```
ip tcp synwait-time 5
```

```
!  
!  
!  
!  
!
```

```
interface FastEthernet0/0
```

```
description *** Unused for Layer2 EtherSwitch ***
```

```
no ip address
```

```
shutdown
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet0/1
```

```
description *** Unused for Layer2 EtherSwitch ***
```

```
ip address 10.2.0.2 255.255.255.0
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface FastEthernet1/0
```

```
duplex full
```

```
speed 100
```

```
!
```

```
interface FastEthernet1/1
```

```
switchport access vlan 22
```

```
duplex full
speed 100
!
interface FastEthernet1/2
switchport access vlan 32
duplex full
speed 100
!
interface FastEthernet1/3
duplex full
speed 100
!
interface FastEthernet1/4
duplex full
speed 100
!
interface FastEthernet1/5
duplex full
speed 100
!
interface FastEthernet1/6
duplex full
speed 100
!
interface FastEthernet1/7
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
duplex full
speed 100
!
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
```

```
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
ip address 10.0.1.4 255.255.255.0
no autostate
!
interface Vlan22
ip address 10.0.22.1 255.255.255.0
no autostate
!
interface Vlan32
ip address 10.0.32.1 255.255.255.0
no autostate
!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
ip forward-protocol nd
!
!
```

```
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
banner exec ^C
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```
^C
!  
line con 0  
exec-timeout 0 0  
privilege level 15  
logging synchronous  
line aux 0  
exec-timeout 0 0  
privilege level 15  
logging synchronous  
line vty 0 4  
login
```

!
!
end

```
ESW3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 7 subnets
C       10.2.0.0 is directly connected, FastEthernet0/1
R       10.1.0.0 [120/1] via 10.2.0.1, 00:00:05, FastEthernet0/1
C       10.0.1.0 is directly connected, Vlan1
R       10.0.31.0 [120/2] via 10.2.0.1, 00:00:05, FastEthernet0/1
C       10.0.22.0 is directly connected, Vlan22
R       10.0.21.0 [120/2] via 10.2.0.1, 00:00:05, FastEthernet0/1
C       10.0.32.0 is directly connected, Vlan32
```

Conectividade e pacotes:

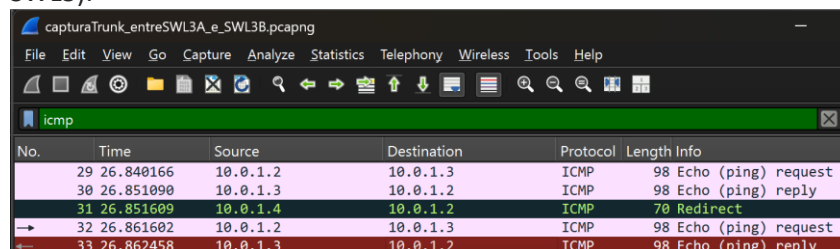
Há conectividade entre as estações da VLAN1:

```
root@Ubu-1:~# ping 10.0.1.3
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data.
64 bytes from 10.0.1.3: icmp_seq=1 ttl=64 time=18.9 ms
From 10.0.1.1 icmp_seq=1 Redirect Host(New nexthop: 10.0.1.3)
From 10.0.1.4 icmp_seq=1 Redirect Host(New nexthop: 10.0.1.3)
64 bytes from 10.0.1.3: icmp_seq=2 ttl=64 time=4.32 ms
^C
--- 10.0.1.3 ping statistics ---
2 packets transmitted, 2 received, +2 errors, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 4.323/11.632/18.941/7.309 ms
root@Ubu-1:~# ping 10.0.1.2
PING 10.0.1.2 (10.0.1.2) 56(84) bytes of data.
64 bytes from 10.0.1.2: icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from 10.0.1.2: icmp_seq=2 ttl=64 time=0.058 ms
64 bytes from 10.0.1.2: icmp_seq=3 ttl=64 time=0.056 ms
^C
--- 10.0.1.2 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2073ms
rtt min/avg/max/mdev = 0.034/0.049/0.058/0.010 ms
```

Há conectividade total entre a VLAN1 e as restantes VLANs (VLANs 21, 31, 22 e 32):

```
root@Ubu-1:~# ping 10.0.21.2
PING 10.0.21.2 (10.0.21.2) 56(84) bytes of data.
64 bytes from 10.0.21.2: icmp_seq=2 ttl=63 time=31.1 ms
64 bytes from 10.0.21.2: icmp_seq=3 ttl=63 time=25.5 ms
64 bytes from 10.0.21.2: icmp_seq=4 ttl=63 time=26.5 ms
64 bytes from 10.0.21.2: icmp_seq=5 ttl=63 time=26.2 ms
64 bytes from 10.0.21.2: icmp_seq=6 ttl=63 time=26.8 ms
^C
--- 10.0.21.2 ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5034ms
rtt min/avg/max/mdev = 25.509/27.211/31.086/1.981 ms
root@Ubu-1:~# ping 10.0.31.2
PING 10.0.31.2 (10.0.31.2) 56(84) bytes of data.
64 bytes from 10.0.31.2: icmp_seq=2 ttl=63 time=17.3 ms
64 bytes from 10.0.31.2: icmp_seq=3 ttl=63 time=25.9 ms
64 bytes from 10.0.31.2: icmp_seq=4 ttl=63 time=19.8 ms
^C
--- 10.0.31.2 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3069ms
rtt min/avg/max/mdev = 17.342/21.035/25.933/3.609 ms
root@Ubu-1:~# ping 10.0.22.2
PING 10.0.22.2 (10.0.22.2) 56(84) bytes of data.
64 bytes from 10.0.22.2: icmp_seq=3 ttl=63 time=28.1 ms
64 bytes from 10.0.22.2: icmp_seq=4 ttl=63 time=29.9 ms
64 bytes from 10.0.22.2: icmp_seq=5 ttl=63 time=32.6 ms
^C
--- 10.0.22.2 ping statistics ---
5 packets transmitted, 3 received, 40% packet loss, time 4030ms
rtt min/avg/max/mdev = 28.102/30.188/32.598/1.849 ms
root@Ubu-1:~# ping 10.0.32.2
PING 10.0.32.2 (10.0.32.2) 56(84) bytes of data.
64 bytes from 10.0.32.2: icmp_seq=2 ttl=63 time=33.5 ms
64 bytes from 10.0.32.2: icmp_seq=3 ttl=63 time=31.9 ms
64 bytes from 10.0.32.2: icmp_seq=4 ttl=63 time=39.9 ms
^C
--- 10.0.32.2 ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3036ms
rtt min/avg/max/mdev = 31.882/35.090/39.862/3.440 ms
SWL3A#ping 10.0.22.1 source 10.0.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.22.1, timeout is 2 seconds:
Packet sent with a source address of 10.0.1.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 24/28/32 ms
SWL3A#
```

A VLAN 1 é ent-to-end, tem conectividade através dos trunks com ela mesma (devido ao comando 'switchport trunk allowed vlan 1,1002-1005'). Mas a conectividade entre a VLAN 1 e as VLANs 21, 31, 22 e 32 é feita através dos links Layer 3 e do routing por eles efetuado (interfaces f0/0 e f0/1 dos 3 SWL3).

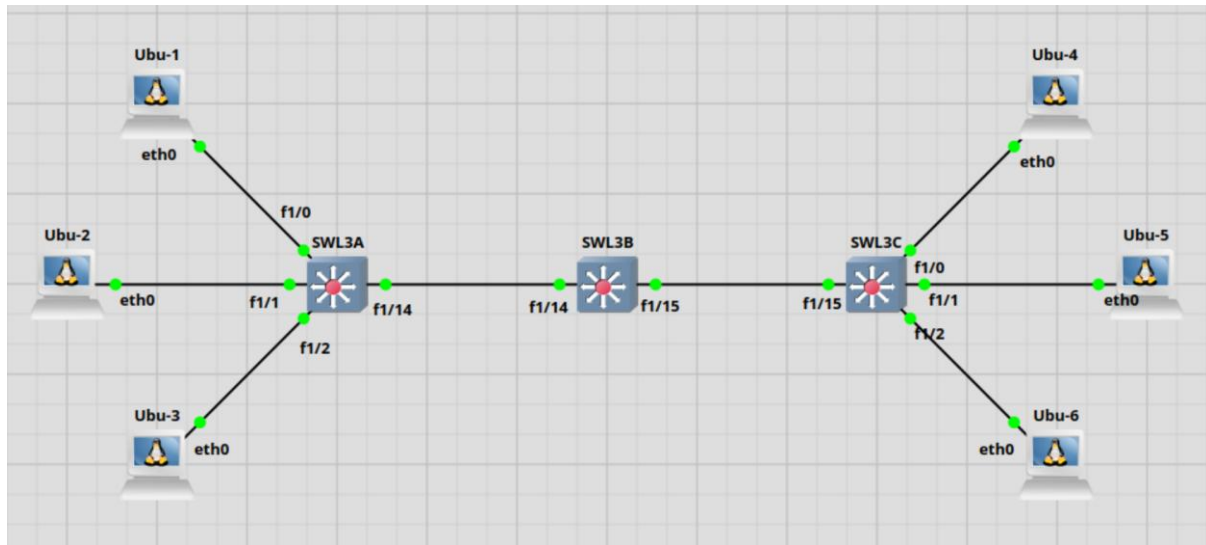


The image shows a Wireshark packet capture window titled 'capturaTrunk_entreSWL3A_e_SWL3B.pcapng'. The filter is set to 'icmp'. The packet list shows five packets:

No.	Time	Source	Destination	Protocol	Length	Info
29	26.840166	10.0.1.2	10.0.1.3	ICMP	98	Echo (ping) request
30	26.851090	10.0.1.3	10.0.1.2	ICMP	98	Echo (ping) reply
31	26.851609	10.0.1.4	10.0.1.2	ICMP	70	Redirect
32	26.861602	10.0.1.2	10.0.1.3	ICMP	98	Echo (ping) request
33	26.862458	10.0.1.3	10.0.1.2	ICMP	98	Echo (ping) reply

Links de Trunk Restringidos (sem Link L3)

9.



```
SWL3A#vlan database
SWL3A(vlan)#vlan 1
VLAN 1 modified:
SWL3A(vlan)#vlan 21
VLAN 21 modified:
SWL3A(vlan)#vlan 31
VLAN 31 modified:
SWL3A(vlan)#vlan 101
VLAN 101 added:
    Name: VLAN0101
SWL3A(vlan)#exit
APPLY completed.
Exiting...
SWL3A#wr

SWL3B#vlan database
SWL3B(vlan)#vlan 1
VLAN 1 modified:
SWL3B(vlan)#vlan 21
VLAN 21 modified:
SWL3B(vlan)#vlan 31
VLAN 31 modified:
SWL3B(vlan)#vlan 22
VLAN 22 modified:
SWL3B(vlan)#vlan 32
VLAN 32 modified:
SWL3B(vlan)#vlan 101
VLAN 101 added:
    Name: VLAN0101
SWL3B(vlan)#exit
APPLY completed.
Exiting...
SWL3B#wr

SWL3C#vlan database
SWL3C(vlan)#vlan 1
VLAN 1 modified:
SWL3C(vlan)#vlan 22
VLAN 22 modified:
SWL3C(vlan)#vlan 32
VLAN 32 modified:
SWL3C(vlan)#vlan 101
VLAN 101 added:
    Name: VLAN0101
SWL3C(vlan)#exit
APPLY completed.
Exiting...
SWL3C#wr
```

Nos três Switches L3, adicionar a vlan 101 às allowed vlans:

```
SWL3A(config)#int range f1/14 - 15
SWL3A(config-if-range)#shut
SWL3A(config-if-range)#
*Mar 1 00:14:03.959: %DTP-5-NONTRUNKPORTON: Port Fa1/14 has become no
SWL3A(config-if-range)#sw
SWL3A(config-if-range)#switchport mod
*Mar 1 00:14:05.443: %LINK-5-CHANGED: Interface FastEthernet1/14, cha
stratively down
*Mar 1 00:14:05.467: %LINK-5-CHANGED: Interface FastEthernet1/15, cha
stratively down
*Mar 1 00:14:06.443: %LINEPROTO-5-UPDOWN: Line protocol on Interface
anged state to down
SWL3A(config-if-range)#switchport mod
SWL3A(config-if-range)#switchport mode tr
SWL3A(config-if-range)#switchport mode trunk
SWL3A(config-if-range)#switchport trunk allowed vlan 1,101,1002-1005
SWL3A(config-if-range)#
SWL3A(config-if-range)#no shut
```

Criar interface virtual vlan 101 e atribuir IP em cada SWL3:

```
SWL3A#conf t
Enter configuration commands, one per line. End with CNTL/Z
SWL3A(config)#interface vlan 101
SWL3A(config-if)#ip add
SWL3A(config-if)#ip address
*Mar 1 00:20:03.647: %LINEPROTO-5-UPDOWN: Line protocol on
te to up
SWL3A(config-if)#ip address 10.0.101.1 255.255.255.0
SWL3A(config-if)#no shut
SWL3B#conf t
Enter configuration commands, one per line. End with
SWL3B(config)#interf
SWL3B(config)#interf
SWL3B(config)#interface vlan 101
SWL3B(config-if)#ip add
SWL3B(config-if)#ip address
*Mar 1 00:21:56.843: %LINEPROTO-5-UPDOWN: Line protoco
te to up
SWL3B(config-if)#ip address 10.0.101.2 255.255.255.0
SWL3B(config-if)#no shut
SWL3C#conf t
Enter configuration commands, one per line. End with C
SWL3C(config)#inter
SWL3C(config)#interface vlan 101
SWL3C(config-if)#
*Mar 1 00:22:28.851: %LINEPROTO-5-UPDOWN: Line protoco
te to up
SWL3C(config-if)#ip
SWL3C(config-if)#ip
SWL3C(config-if)#ip add
SWL3C(config-if)#ip address 10.0.101.3 255.255.255.0
SWL3C(config-if)#no shut
```

Fazer 'ip routing' de novo em cada SWL3 em global configuration mode para tornar possível o inter-vlan routing/ routing entre VLANs diferentes (agora incluindo a vlan 101).

Tabelas de encaminhamento dos SWL3:

```
SWL3A#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 6 subnets
C       10.0.1.0 is directly connected, Vlan1
C       10.0.31.0 is directly connected, Vlan31
R       10.0.22.0 [120/1] via 10.0.101.3, 00:00:16, Vlan101
C       10.0.21.0 is directly connected, Vlan21
R       10.0.32.0 [120/1] via 10.0.101.3, 00:00:16, Vlan101
C       10.0.101.0 is directly connected, Vlan101

SWL3B#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 6 subnets
R       10.0.1.0 [120/1] via 10.0.101.3, 00:00:22, Vlan101
           [120/1] via 10.0.101.1, 00:00:20, Vlan101
R       10.0.31.0 [120/1] via 10.0.101.1, 00:00:20, Vlan101
R       10.0.22.0 [120/1] via 10.0.101.3, 00:00:22, Vlan101
R       10.0.21.0 [120/1] via 10.0.101.1, 00:00:20, Vlan101
R       10.0.32.0 [120/1] via 10.0.101.3, 00:00:22, Vlan101
C       10.0.101.0 is directly connected, Vlan101
```

```

SWL3C#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 6 subnets
C       10.0.1.0 is directly connected, Vlan1
R       10.0.31.0 [120/1] via 10.0.101.1, 00:00:16, Vlan101
C       10.0.22.0 is directly connected, Vlan22
R       10.0.21.0 [120/1] via 10.0.101.1, 00:00:16, Vlan101
C       10.0.32.0 is directly connected, Vlan32
C       10.0.101.0 is directly connected, Vlan101

```

Conectividade e pacotes:

```

SWL3A#ping 10.0.22.2 source 10.0.21.1

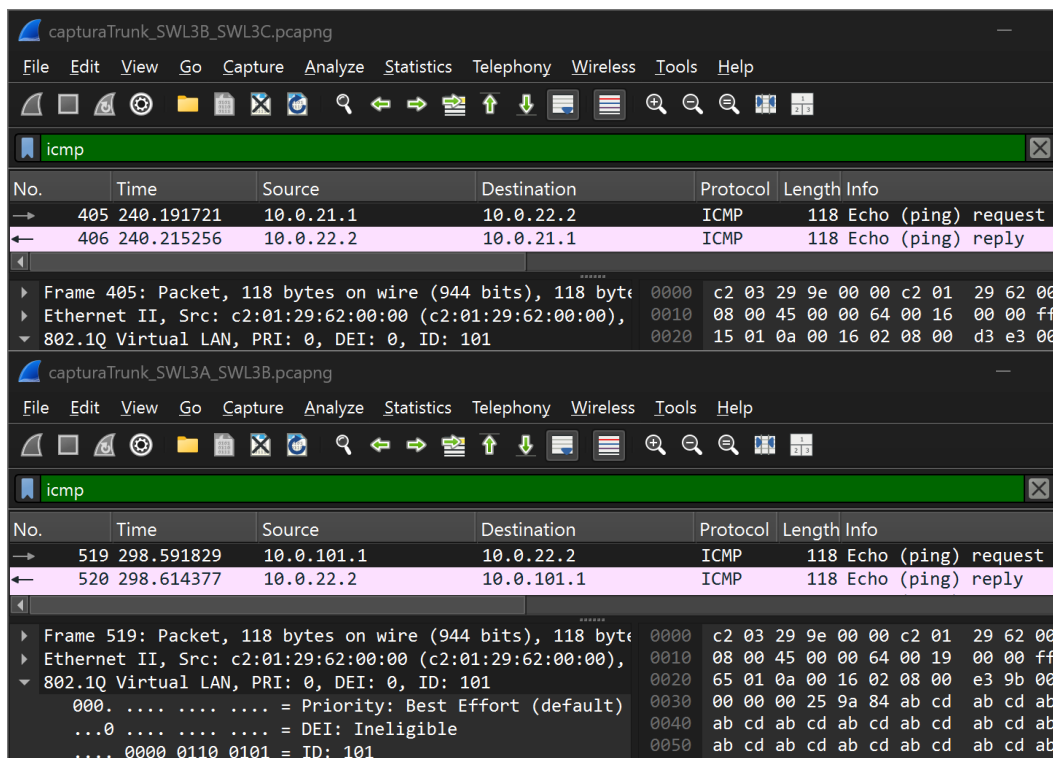
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.22.2, timeout is 2 seconds:
Packet sent with a source address of 10.0.21.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 72/110/148 ms
SWL3A#ping 10.0.22.2 source 10.0.101.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.22.2, timeout is 2 seconds:
Packet sent with a source address of 10.0.101.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/35/48 ms
SWL3A#ping 10.0.32.2 source 10.0.101.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.32.2, timeout is 2 seconds:
Packet sent with a source address of 10.0.101.1
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 24/30/36 ms
SWL3A#ping 10.0.32.2 source 10.0.31.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.32.2, timeout is 2 seconds:
Packet sent with a source address of 10.0.31.1
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/155/220 ms

```



Por exemplo, nestes recortes de capturas nos trunks entre SWL3B e SWL3C e entre SWL3A e SWL3B, respetivamente, é possível verificar que o tráfego L3 inter-VLAN (entre Vlans diferentes) é feito através do trunk e encapsulado na vlan 101 ("ID: 101" numa frame com a tag 802.1Q), em que o RIPv2 agora troca rotas através da VLAN 101 e do trunk e não através de um link L3 como foi feito no exercício 7.

Conclusão: apesar de não haver links de Layer 3, a vlan 101 (de interligação) anuncia as redes 10.0.21.0, 10.0.22.0, 10.0.31.0 e 10.0.32.0 das outras VLANs através da VLAN 101. Ou seja, as rotas entre VLANs locais dos Switches L3 passaram a ser trocadas via VLAN 101 sobre os trunks.

Configuração SWL3A:

```
SWL3A#sh run
Building configuration...
```

```
Current configuration : 3058 bytes
```

```
!  
version 12.4  
service timestamps debug datetime msec  
service timestamps log datetime msec  
no service password-encryption  
no service dhcp  
!  
hostname SWL3A  
!  
boot-start-marker  
boot-end-marker  
!  
!  
no aaa new-model  
memory-size iomem 5  
no ip icmp rate-limit unreachable  
ip cef  
!  
!  
!  
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***
```

```
ip address 10.1.0.2 255.255.255.0
shutdown
duplex auto
speed auto
!
interface FastEthernet0/1
description *** Unused for Layer2 EtherSwitch ***
no ip address
shutdown
duplex auto
speed auto
!
interface FastEthernet1/0
duplex full
speed 100
!
interface FastEthernet1/1
switchport access vlan 21
duplex full
speed 100
!
interface FastEthernet1/2
switchport access vlan 31
duplex full
speed 100
!
interface FastEthernet1/3
duplex full
speed 100
!
interface FastEthernet1/4
duplex full
speed 100
!
interface FastEthernet1/5
duplex full
speed 100
!
interface FastEthernet1/6
duplex full
speed 100
!
interface FastEthernet1/7
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
```

```
duplex full
speed 100
!
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
ip address 10.0.1.1 255.255.255.0
no autostate
!
interface Vlan21
ip address 10.0.21.1 255.255.255.0
no autostate
!
interface Vlan31
ip address 10.0.31.1 255.255.255.0
no autostate
!
interface Vlan101
ip address 10.0.101.1 255.255.255.0
```

```

!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C

```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```

^C
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15

```

```
logging synchronous
line vty 0 4
login
!
!
end
```

Configuração SWL3B:

```
SWL3B#sh run
Building configuration...
```

```
Current configuration : 2862 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname SWL3B
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
!
no ip domain lookup
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
!
!
!
!
!
!
!
!
!
```

```
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.2.0.1 255.255.255.0  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.1.0.1 255.255.255.0  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100  
!  
interface FastEthernet1/1  
duplex full  
speed 100  
!  
interface FastEthernet1/2  
duplex full  
speed 100  
!  
interface FastEthernet1/3  
duplex full  
speed 100  
!  
interface FastEthernet1/4  
duplex full  
speed 100  
!  
interface FastEthernet1/5  
duplex full  
speed 100  
!  
interface FastEthernet1/6
```

```
duplex full
speed 100
!
interface FastEthernet1/7
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
duplex full
speed 100
!
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
duplex auto
speed auto
!
interface Vlan1
no ip address
```

```

shutdown
!
interface Vlan101
ip address 10.0.101.2 255.255.255.0
!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C

```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

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Please update the IOS to enable the "macro" command!

```

^C
!
line con 0
exec-timeout 0 0
privilege level 15

```

```
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end
```

Configuração SWL3C:

```
SWL3C#sh run
Building configuration...
```

```
Current configuration : 3058 bytes
!
version 12.4
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
no service dhcp
!
hostname SWL3C
!
boot-start-marker
boot-end-marker
!
!
no aaa new-model
memory-size iomem 5
no ip icmp rate-limit unreachable
ip cef
!
!
!
!
no ip domain lookup
ip auth-proxy max-nodata-conns 3
ip admission max-nodata-conns 3
!
!
!
!
!
!
!
!
```

```
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
ip address 10.2.0.2 255.255.255.0  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100  
!  
interface FastEthernet1/1  
switchport access vlan 22  
duplex full  
speed 100  
!  
interface FastEthernet1/2  
switchport access vlan 32  
duplex full  
speed 100  
!  
interface FastEthernet1/3  
duplex full  
speed 100  
!  
interface FastEthernet1/4  
duplex full  
speed 100  
!
```

```
interface FastEthernet1/5
duplex full
speed 100
!
interface FastEthernet1/6
duplex full
speed 100
!
interface FastEthernet1/7
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
duplex full
speed 100
!
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/15
switchport trunk allowed vlan 1,101,1002-1005
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet2/0
no ip address
shutdown
```

```

duplex auto
speed auto
!
interface Vlan1
ip address 10.0.1.4 255.255.255.0
no autostate
!
interface Vlan22
ip address 10.0.22.1 255.255.255.0
no autostate
!
interface Vlan32
ip address 10.0.32.1 255.255.255.0
no autostate
!
interface Vlan101
ip address 10.0.101.3 255.255.255.0
!
router rip
version 2
passive-interface Vlan1
network 10.0.0.0
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
!
banner exec ^C

```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode

After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

^C

!

line con 0

exec-timeout 0 0

privilege level 15

logging synchronous

line aux 0

exec-timeout 0 0

privilege level 15

logging synchronous

line vty 0 4

login

!

!

end